

Plant Production vs. Our AI Plan

Pattern Analysis & KPI Benchmark — May 2026 (JK Tyre BTP PCR Line)

Source: plant Stage-1 & Stage-2 production logs (production_stage1_as.csv, production_stage2_as.csv) — the full GT-building window for May demand (Apr 27 → May 31; the plant pre-builds ~4 days early so GT is ready to cure on time, mirroring our scheduler's pre-start).
Benchmark: demand_may.xlsx (694k, 85 SKUs). Our AI Plan: B2C rolling scheduler, max curing changeovers = **18/day**. Scope: the 79 SKUs in BOTH demand and production; the 168 other plant SKUs (39,490 tyres, 5% of GT volume) are **not in demand** and are excluded.

Executive Summary — the headline

Our AI Plan already matches/beats the plant on useful output, with essentially zero waste. The plant appears to produce more (727,069 GT for our demand SKUs = 104.8% of demand), but 61,453 of those tyres are **overbuild beyond demand** — physically wasted, since a SKU cannot be fulfilled beyond its own demand. On a demand-capped (effective) basis the plant delivers **95.9%**; our AI Plan delivers **668,937 cured (96.4%)** while building 64k fewer tyres and wasting none.

Metric	Plant (May actual)	Our AI Plan	Verdict
GT built for demand SKUs	727,069 (104.8%)	662,600 (95.5%)	—
Final cured (fulfilment)	(curing logs N/A)	668,937 (96.4%)	—
Wasted overbuild (> demand)	61,453	0	AI Plan wins
Effective fulfilment (demand-capped)	95.9%	96.4% cured	AI Plan wins
Unmet demand (underbuild)	28,132	~24,800	Comparable
Max curing changeovers / day	(building ~271/day)	18 (config)	Different strategy

Bottom line: raw tyre count is the wrong yardstick — the plant's extra volume is mostly waste. Measured on what actually counts (demand fulfilled), we are already ahead. The remaining gap is **structural**, and the plant hits it too.

1. The plant over-builds runners and under-builds others (mis-allocation)

The plant does not respect a strict per-SKU demand cap. It over-built **41** of our demand SKUs and under-built **18**. It pours extra capacity into high-volume runners (+10% each) while leaving other SKUs largely unserved:

Over-built runner (SKU)	Demand	Plant built	%	Under-built (SKU)	Demand	Plant built	%
1325214813075SUNE1	70,205	77,303	110%	1325215813079TTMX0	12,521	104	1%
1225170015012LSTL0	69,472	76,531	110%	1325217613082TUNE0	12,494	3,308	26%
1325218515088HTMX0	33,144	36,796	111%	1325218515088HURL0	1,847	0	0%
1D25213812090FXPC0	25,309	27,757	110%	1325221416095VURSO	1,242	0	0%

Our AI Plan's advantage: it enforces the demand cap (build ≤ demand, now with zero overbuild after this month's fix) and allocates by real deficit/ratio. That is exactly what converts the plant's 61k of wasted overbuild into either fulfilment of under-served SKUs or genuine capacity savings.

2. Our AI Plan corrects the plant's mis-allocation (serves the SKUs the plant abandoned)

The most striking result: the very SKUs the plant left on the table, our AI Plan serves **fully**. While the plant poured capacity into overbuilding runners, it abandoned ~28k units of real demand on other SKUs — our demand-aware allocation redistributes to them instead:

SKU	Demand	Plant built	Plant %	AI Plan cured	AI Plan %
1325215813079TTMX0	12,521	104	1%	12,521	100%
1325217613082TUNE0	12,494	3,308	26%	12,462	100%
1325218515088HURL0	1,847	0	0%	1,800	97%
1325221416095VURS0	1,242	0	0%	1,232	99%

Interpretation. Both the plant and our AI Plan land near ~96% overall — the aggregate ~4% gap is a genuine physical capacity limit (press count, 15" tooling). But the two spend that 96% very differently: the plant wastes 61k overbuilding runners while skipping large-demand SKUs like TTMX0/TUNE0; our AI Plan wastes nothing and keeps every major-demand SKU served. **Same ceiling, far smarter allocation.**

3. How the plant actually runs — patterns we don't replicate

(a) High-changeover, generalist strategy. Plant machines switch SKU constantly — the opposite of our CO-minimising design:

Changeover / campaign metric	Plant (actual)	Our AI Plan
Building COs per day (whole fleet)	271 avg / 517 max	far lower (CO-minimising)
Curing changeovers per day (cap)	(building ~271/day)	18 / day (config)
COs per machine per day	~10 (up to 826/month on vmi2Maxx)	≤ 6 (max 2 building CO/shift)
Avg campaign length before switch	~83 tyres (~1.5 h)	long (MIN_CAMPAIGN 60 min, few COs)
Same-inch vs diff-inch COs	82% same / 18% diff	heavily avoids diff-inch

(b) VMI machines: plant runs them across many inches; our AI Plan locks them to one. This is the biggest structural divergence. Plant VMI machines each build 38–50 SKUs spanning **4–7 different inches**. Our AI Plan hard-locks each VMI machine to its **1 dominant inch** (it still builds several SKUs, but all of one size). Side by side:

Plant VMI	SKUs	Inches	Our VMI	SKUs	Inches
vmi2Maxx	42	7 (12–18)	7003	20	3 (14–16)
vmi3Maxx	50	5 (13–17)	7001	16	2 (15–16)
vmi1Maxx	39	5 (13–18)	7004	13	1 (14)
VMIExxium01	38	4 (14–17)	6001	10	1 (14)

The gap is **inch range**, not SKU count: our VMI machines already build many SKUs — but all within one locked size. The plant freely mixes 4–7 inches on the same VMI machine.

(c) Where GT gets built — by machine group, plant vs AI Plan (demand SKUs). Because our inch-locking pushes 15" work off VMI, our AI Plan leans more on BJ (34% vs the plant's 28%) and less on VMI (42% vs 47%):

Machine group	Plant GT	Plant %	AI Plan GT	AI Plan %
VMI	341,114	47%	275,797	42%
BJ	201,122	28%	222,829	34%
IRM / Two-Stage	135,409	19%	126,364	19%
UNISTAGE	49,424	7%	37,610	6%
Total	727,069	100%	662,600	100%

4. Can we target better KPIs than the plant?

Yes — and on the metric that matters, we already do.

- **On efficiency (fulfilment per tyre built): already winning.** Our AI Plan cures 668,937 (96.4%) with zero waste vs the plant's 95.9% effective with 61k wasted overbuild.
- **On raw volume: the plant builds more, but it is mostly waste.** Matching 727k is not a goal — 61k of it fulfils nothing.
- **To push past 96.4%: our residual ~25k unmet is on curing-throughput-limited SKUs** — a *different* set from the ones the plant skipped (which our AI Plan already serves fully). Closing it needs physical curing-press capacity for the 15"-oversubscribed group, not a schedule change — add that capacity and our demand-aware allocation pulls clearly ahead of the plant.

5. Suggestions to optimise — with proof & expected KPI change

Every row below is backed by measured data. Baseline for "expected change" is our current AI Plan: **668,937 cured (96.4%)** at max curing changeovers = 18/day.

#	Change	Proof (measured evidence)	Expected KPI	Effort / Risk
1	Raise curing changeover cap 18 → 26 / day	Tested sweep (this run): 18 → 668,937 (96.4%) 22 → 669,919 (96.6%) 26 → 671,058 (96.7%) 30 → 671,472 (96.8%)	+2,100 cured → 96.7% (diminishing past 26)	1-line config Low risk
2	Multi-inch VMI (adopt plant flexibility) + high changeover, as a paired pilot	Plant VMI builds 38–50 SKUs / 4–7 inches per machine; ours locked to 1–3. Plant pairs it with ~10 CO/machine/day, which prevents lock-in.	Potential +5k to +15k IF carry-over solved. Unproven — pilot	High effort Med-High risk
3	Add curing presses for the 15"-oversubscribed group (physical / CAPEX)	Our ~24.8k unmet is 15"-curing-limited; plant leaves ~28k unmet too, even at 4× our changeover rate — the same physical ceiling.	Up to +24.8k → ~99% (not a scheduler change)	Physical CAPEX
4	Keep delivered fixes (overbuild cap + building ratio) — do not regress	Overbuild 3,303 → 29 units; GT write-off 2,962 → 667; building-ratio symmetry +1,169 cured. All measured.	Locked in (already in the 96.4% baseline)	None (preserve)

Recommended sequence. (1) Ship the CO-cap raise to 26 now — it is a one-line, low-risk +0.3pp (96.4% → 96.7%). (2) Scope suggestion 2 as a proper pilot: the plant proves multi-inch VMI works, but only *with* its high changeover rate — earlier isolated inch-relaxation tests regressed precisely because they lacked that pairing (machines locked onto off-inch SKUs). Test the two together, behind a toggle, with return-to-dominant logic. (3) Feed suggestion 3 (15" press capacity) to plant planning — it is the only lever that lifts us clearly past ~96.5%, and the plant data proves the gap is physical, not algorithmic.

Data caveats: production RecipeCode is a text description; ~95% of GT volume mapped to our SKU codes via the building-event dictionary, the remaining ~5% are low-volume non-demand SKUs. Plant figures are GT *built* (Stage-2 production); plant *curing* logs were not provided, so the head-to-head uses plant effective-built vs our cured/built. Changeovers are counted as SKU transitions in time-ordered per-machine event streams.